# Docker in ubuntu

**Follow the step:** (using the apt repository)

Step-1: Add Docker's official GPG key:

# sudo apt update

# sudo apt install ca-certificates curl

# sudo install -m 0755 -d /etc/apt/keyrings

# sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc

# sudo chmod a+r /etc/apt/keyrings/docker.asc

```
siddharrth@master:~$ sudo apt update
[sudo] password for siddharrth:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 https://packages.gitlab.com/gitlab/gitlab-ee/ubuntu/noble noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

```
siddharrth@master:~$ sudo apt install ca-certificates curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
curl is already the newest version (8.5.0-2ubuntu10.9).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

```
siddharrth@master:~$ sudo install -m 0755 -d /etc/apt/keyrings
[sudo] password for siddharrth:
siddharrth@master:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
siddharrth@master:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
```

Step-2: Add the repository to Apt sources:

# sudo tee /etc/apt/sources.list.d/docker.sources <<EOF

Types: deb

URIs: https://download.docker.com/linux/ubuntu

Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")

Components: stable

Architectures: $(dpkg --print-architecture)

Signed-By: /etc/apt/keyrings/docker.asc

EOF

# sudo apt update

```
siddharrth@master:~$ sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: noble
Components: stable
Architectures: amd64
Signed-By: /etc/apt/keyrings/docker.asc
siddharrth@master:~$ sudo apt update
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:2 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [53.4 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:7 https://packages.gitlab.com/gitlab/gitlab-ee/ubuntu/noble noble InRelease
Fetched 102 kB in 1s (81.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

Step-3: Install the Docker packages.

# sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

```
siddharrth@master:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras libslirp0 pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libslirp0
  pigz slirp4netns
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 94.9 MB of archives.
After this operation, 365 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 2.2.3-1~ubuntu.24.04~noble [23.6 M
B]
Get:2 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble/main amd64 libslirp0 amd64 4.7.0-1ubuntu3 [63.8 kB]
```

Step-4: Check the docker status

# docker –version

```
siddharrth@master:~$ docker --version
Docker version 29.4.3, build 055a478
```

# sudo systemctl status docker

```
siddharrth@master:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-05-11 12:17:28 UTC; 9min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2418 (dockerd)
      Tasks: 9
     Memory: 27.7M (peak: 28.7M)
        CPU: 626ms
     CGroup: /system.slice/docker.service
             └─2418 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

May 11 12:17:27 master dockerd[2418]: time="2026-05-11T12:17:27.421036733Z" level=info msg="Restoring containers: start>
May 11 12:17:27 master dockerd[2418]: time="2026-05-11T12:17:27.536731437Z" level=info msg="Deleting nftables IPv4 rule>
May 11 12:17:27 master dockerd[2418]: time="2026-05-11T12:17:27.547647168Z" level=info msg="Deleting nftables IPv6 rule>
May 11 12:17:28 master dockerd[2418]: time="2026-05-11T12:17:28.427078623Z" level=info msg="Loading containers: done."
May 11 12:17:28 master dockerd[2418]: time="2026-05-11T12:17:28.438619203Z" level=info msg="Docker daemon" commit=56be7>
May 11 12:17:28 master dockerd[2418]: time="2026-05-11T12:17:28.438857512Z" level=info msg="Initializing buildkit"
May 11 12:17:28 master dockerd[2418]: time="2026-05-11T12:17:28.624533270Z" level=info msg="Completed buildkit initiali>
May 11 12:17:28 master dockerd[2418]: time="2026-05-11T12:17:28.642298512Z" level=info msg="Daemon has completed initia>
May 11 12:17:28 master dockerd[2418]: time="2026-05-11T12:17:28.642971925Z" level=info msg="API listen on /run/docker.s>
May 11 12:17:28 master systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
```

Step-5: Verify that the installation is successful by running the hello-world image

```
siddharrth@master:~$  sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

Some common commands

List containers

# docker ps

```
sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

## List all containers

# docker ps –a

```
siddharrth@master:~$ sudo docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED         STATUS                     PORTS     NAMES
8847fb2754da   hello-world   "/hello"   8 minutes ago   Exited (0) 8 minutes ago             gracious_poitras
```

## List images

# docker images

```
siddharrth@master:~$ sudo docker images
                                                                                    i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   f9078146db2e       25.9kB         9.49kB    U
```

## Run container

# docker run ubuntu

```
sudo docker pull siddhumahi01/httpd:latest
latest: Pulling from siddhumahi01/httpd
589a19d041b4: Pull complete
d2c754535bea: Pull complete
23f5dfc9bc5d: Pull complete
Digest: sha256:6c7eb2c67c89a53adefcf8ed53aac02693db44408797e96cfa23589bb63016af
Status: Downloaded newer image for siddhumahi01/httpd:latest
docker.io/siddhumahi01/httpd:latest
siddharrth@master:~$ sudo docker ps -a
CONTAINER ID   IMAGE         COMMAND       CREATED          STATUS                     PORTS     NAMES
3e43a15169cf   ubuntu        "/bin/bash"   10 minutes ago   Exited (0) 10 minutes ago            lucid_hermann
8847fb2754da   hello-world   "/hello"      20 minutes ago   Exited (0) 20 minutes ago            gracious_poitras
siddharrth@master:~$ sudo docker images
                                                                                    i Info →   U  In Use
IMAGE                       ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest          f9078146db2e       25.9kB         9.49kB    U
siddhumahi01/httpd:latest   6c7eb2c67c89        447MB          131MB
ubuntu:latest               f3d28607ddd7        160MB         45.3MB    U
siddharrth@master:~$
```